OPD internal components: Query Residual x LN history

512 actual complete fits; 12800 recorded rounds; four matched arms; both original development orders

F / B / A = Frozen / Before / After. Absolute values in percent. R30/R50 use strict per-query vIoU > 0.3 / 0.5.

Primary: source-frozen actionness projection I1

Target / arm	F v	B v	A v	F R30	B R30	A R30	F R50	B R50	A R50
HC-STVG-v2 / Joint + carry	29.017	30.065	32.781	46.875	46.875	48.438	12.500	15.625	25.000
HC-STVG-v2 / LN + carry	29.017	30.071	32.761	46.875	48.438	50.000	12.500	15.625	25.000
HC-STVG-v2 / Joint + reset	29.017	29.017	32.415	46.875	46.875	50.000	12.500	12.500	25.000
HC-STVG-v2 / LN + reset	29.017	29.017	32.494	46.875	46.875	50.000	12.500	12.500	25.000
VidSTG / Joint + carry	12.599	13.935	16.822	9.375	15.625	28.125	0.000	3.125	3.125
VidSTG / LN + carry	12.599	14.027	16.460	9.375	17.188	28.125	0.000	3.125	3.125
VidSTG / Joint + reset	12.599	12.599	15.996	9.375	9.375	28.125	0.000	0.000	0.000
VidSTG / LN + reset	12.599	12.599	15.725	9.375	9.375	28.125	0.000	0.000	0.000

Secondary: Native I0, evaluated on the same spatial boxes

Target / arm	F v	B v	A v	F R30	B R30	A R30	F R50	B R50	A R50
HC-STVG-v2 / Joint + carry	24.571	25.426	28.143	34.375	37.500	37.500	9.375	12.500	20.312
HC-STVG-v2 / LN + carry	24.571	25.446	27.921	34.375	39.062	37.500	9.375	14.062	20.312
HC-STVG-v2 / Joint + reset	24.571	24.571	27.942	34.375	34.375	37.500	9.375	9.375	21.875
HC-STVG-v2 / LN + reset	24.571	24.571	27.772	34.375	34.375	37.500	9.375	9.375	21.875
VidSTG / Joint + carry	12.364	13.615	16.389	9.375	15.625	28.125	3.125	3.125	0.000
VidSTG / LN + carry	12.364	13.719	16.085	9.375	17.188	28.125	3.125	3.125	0.000
VidSTG / Joint + reset	12.364	12.364	15.652	9.375	9.375	28.125	3.125	3.125	0.000
VidSTG / LN + reset	12.364	12.364	15.430	9.375	9.375	28.125	3.125	3.125	0.000

QR and Adam reset at every query; LN-only freezes QR at zero (1536 optimizer coordinates); joint uses 1792.

HC: lr .01, sigma .025, tau .05, K40, carry 1/16. Vid: .03 / .1 / .25 / K10, carry 1/8. No parameter search.

Native Uniform4 admitted Top1 cached evidence and original key/round noise; each arm maintains its own state chain.

Six fixed contrasts: paired task effects and every tail

Difference is the specified first structure minus reference; 10000 parent bootstrap resamples, seed 20261013

After mvIoU; both orders averaged within each of 32 parents per target

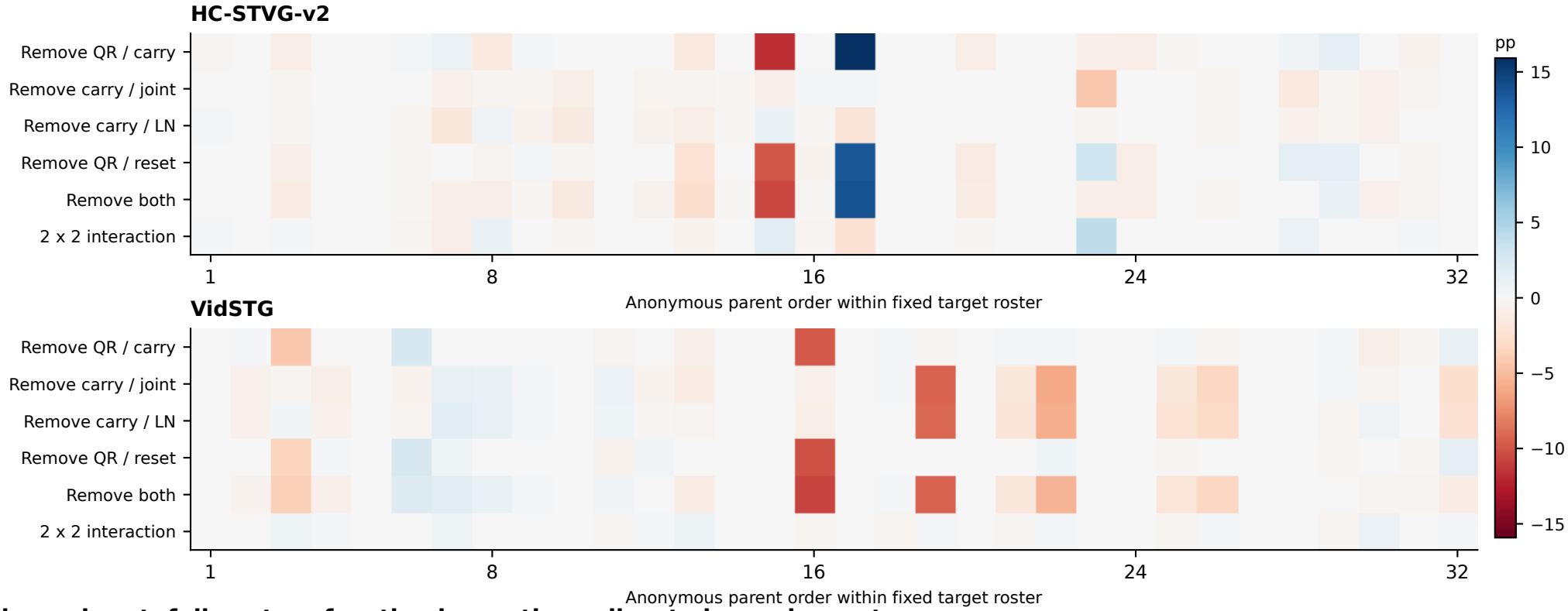
Target	Contrast	I1 delta pp [95% CI]	I0 delta pp [95% CI]	+ / - / 0 / ?	Harm >5/>20 pp	Order1 / order2 pp
HC-STVG-v2	Remove QR / carry	-0.019 [-1.180, +1.345]	-0.222 [-0.996, +0.417]	9/19/4/0	1/0	-0.032/-0.007
HC-STVG-v2	Remove carry / joint	-0.366 [-0.689, -0.145]	-0.202 [-0.334, -0.088]	8/20/4/0	0/0	-0.422/-0.310
HC-STVG-v2	Remove carry / LN	-0.267 [-0.485, -0.068]	-0.149 [-0.311, -0.006]	9/19/4/0	0/0	-0.310/-0.224
HC-STVG-v2	Remove QR / reset	+0.080 [-0.914, +1.224]	-0.169 [-0.774, +0.341]	9/19/4/0	1/0	+0.080/+0.080
HC-STVG-v2	Remove both	-0.286 [-1.306, +0.905]	-0.371 [-1.069, +0.197]	5/23/4/0	1/0	-0.342/-0.230
HC-STVG-v2	2 x 2 interaction	+0.099 [-0.195, +0.450]	+0.053 [-0.118, +0.252]	13/15/4/0	0/0	+0.112/+0.086
VidSTG	Remove QR / carry	-0.362 [-1.122, +0.177]	-0.305 [-1.016, +0.194]	11/11/10/0	1/0	-0.305/-0.419
VidSTG	Remove carry / joint	-0.826 [-1.603, -0.206]	-0.738 [-1.379, -0.200]	7/15/10/0	2/0	-0.578/-1.073
VidSTG	Remove carry / LN	-0.735 [-1.501, -0.126]	-0.655 [-1.296, -0.121]	8/14/10/0	2/0	-0.544/-0.925
VidSTG	Remove QR / reset	-0.271 [-1.053, +0.260]	-0.222 [-0.960, +0.279]	10/7/15/0	1/0	-0.271/-0.271
VidSTG	Remove both	-1.097 [-2.115, -0.261]	-0.960 [-1.865, -0.219]	8/14/10/0	3/0	-0.849/-1.344
VidSTG	2 x 2 interaction	+0.091 [-0.013, +0.204]	+0.083 [-0.023, +0.195]	13/9/10/0	0/0	+0.034/+0.148

Remove QR / carry: LN carry - joint carry; remove carry / joint: joint reset - joint carry.
Remove carry / LN: LN reset - LN carry; remove QR / reset: LN reset - joint reset; remove both: LN reset - joint carry.
Interaction = (LN reset - joint reset) - (LN carry - joint carry). No contrast presumes that deletion succeeds.

All signs and unknowns are retained. Severe harms use strict delta < -0.05 / -0.20, i.e. more than 5 / 20 pp.
These unadjusted exploratory intervals are conditional on historically exposed development sources.
A CI crossing zero is not equivalence. Cost-based simplification needs a predeclared noninferiority margin.
Do not select a different structure for each target domain. The OPD core remains fixed; no production promotion.

All anonymous parent effects and actual instrumented cost

No parent removed; color shows I1 After-mvIoU contrast in percentage points; full parent and order data accompany the PDF



Actual logged cost; full capture function invocations allocated once by capture_owner

Target / arm	Fits	Rounds	Backwards	Empty	Active	Fit sec / fit	Capture total sec	Full capture invocations
HC-STVG-v2 / Joint + carry	64	2560	2480	0	1792	2.3804	284.633	64
HC-STVG-v2 / LN + carry	64	2560	2480	0	1536	2.1241	0.000	0
HC-STVG-v2 / Joint + reset	64	2560	2480	0	1792	2.4231	0.000	0
HC-STVG-v2 / LN + reset	64	2560	2480	0	1536	2.2357	0.000	0
VidSTG / Joint + carry	64	640	460	18	1792	0.5592	241.739	64
VidSTG / LN + carry	64	640	460	18	1536	0.5099	0.000	0
VidSTG / Joint + reset	64	640	460	18	1792	0.5586	0.000	0
VidSTG / LN + reset	64	640	460	18	1536	0.5088	0.000	0

Fit seconds include trace/audit/synchronization and cache reuse; cached teacher generation and serialization are outside this sub-timing.
16 qualification + 128 formal full capture invocations; two source offset forwards each = 32 + 256 source-offset model forwards.
Extra decoder replay calls are outside this counter. Other arms share owner captures; zero allocation does not imply free video access.
Before is each arm own incoming state. After-Before: current effect; Before-Frozen: inherited effect; After-Frozen: total.